

Descriptor: *Comprehensive Multi-Technology PV Operational and Environmental Characterization Dataset*

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ABSTRACT The rapid expansion of photovoltaic (PV) solar energy has motivated the use of data-driven and artificial intelligence (AI) methods to address challenges related to performance modeling, monitoring, and control. However, the efficacy of AI in this domain is often constrained by the scarcity of high-fidelity datasets capturing long-term outdoor operations. To address this, we present a comprehensive dataset comprising over 1.7 million samples collected over four years at an outdoor solar laboratory in Lima, Peru (12°04' S, 77°04' W). The data synchronize PV performance and ambient measurements, including electrical operational data (voltage, current, power), electrical characteristics (open-circuit voltage, short-circuit current, series and shunt resistances), and meteorological parameters (irradiance at different orientations, wind speed and direction, temperature, humidity, etc.). This multi-technology, high-granularity dataset is designed to train and benchmark AI-driven algorithms, providing a validated foundation for the development of robust, technology-agnostic models for real-time energy management and fault diagnosis.

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DATA TYPE/LOCATION Time-series; Outdoor Photovoltaic Research Laboratory, Pontificia Universidad Católica del Perú, Lima, Peru

INDEX TERMS AI, Ambient Conditions, Machine Learning, Outdoor I-V curve, Photovoltaic Systems, Renewable Energy, Solar Power Generation

BACKGROUND

Existing datasets for solar energy focus on a limited set of variables [1], a short span of time (hours to a few days) [2], [3], or lack on-site measurements [4]. These constraints limit the generalizability of AI models in real-world deployments, as the data fails to capture the seasonality of environmental variables and the diversity of PV technologies.

The presented dataset provides outdoor measurements of multiple PV module technologies, along with synchronized environmental and operational parameters. Electrical variables are derived from measured current–voltage (I–V) curves and include voltage, current, and power at the maxi-

mum power point, short-circuit current, open-circuit voltage, series and shunt resistances, and fill factor. Environmental measurements include irradiance under multiple orientations (horizontal, tilted, vertical, and reflected), module temperature, ambient temperature, relative humidity, air density, absolute air pressure, and wind speed and direction.

Consequently, this dataset addresses the data limitation problem by offering a record of multiple PV technologies, providing significantly greater temporal breadth and variable diversity than existing, comparable datasets [5].

The dataset is intended to support AI model training, benchmarking, and validation for power and energy system

applications, including PV modeling, performance prediction, PV forecasting, fault detection, and maximum power point tracking (MPPT).

This particular dataset has been used in past work related to the characterization of photovoltaic panels of different technologies (material characterization) [6]–[8], spectral distribution evaluation [9], [10], as well as for quality control (performance monitoring compared to manufacturers' specifications) [11], [12]. Combined with other electroluminescence imaging systems, this data allows for a precise assessment of the physical, mechanical, and electrical condition of solar cells and panels in an installation [13].

COLLECTION METHODS AND DESIGN

We collected experimental data at the Outdoor Photovoltaic Research Laboratory of the Pontificia Universidad Católica del Perú (PUCP), located in the coastal area of Lima, Peru ($12^{\circ}04' \text{ S}$, $77^{\circ}04' \text{ W}$). The facility is installed on the rooftop of the Physics Department building.

The system comprises a custom-built capacitive-load I–V curve tracer synchronized with a comprehensive monitoring system [12], which records plane-of-array irradiance, spectral distribution, module temperature, and relevant meteorological variables. This configuration enables simultaneous electrical and environmental characterization of the PV modules under outdoor operating conditions.

Monofacial PV modules were mounted at a 20° north-facing tilt angle, as shown in Fig. 1(a). Bifacial PV modules were installed on two different supporting structures specifically designed for bifacial operation, as described in [8]. One structure consists of a 15° north-facing tilted configuration (Fig. 1(b)), while the second corresponds to a vertical east–west configuration with the front side facing west (Fig. 1(c)).

The laboratory is capable of sequentially measuring the current–voltage (I–V) curves of up to 15 PV modules, with a 5-minute interval assigned to each module. Simultaneously, irradiance on the different planes of incidence is recorded together with its spectral distribution, module temperature, and relevant meteorological variables affecting PV performance.

The solar spectrum is measured using an EKO MS-711 spectroradiometer (300–1100 nm), installed at the same tilt angle as the PV modules. Irradiance is measured with EKO MS-80 pyranometers (0–4000 W/m^2). One pyranometer is mounted horizontally, and another at 20° for the monofacial station. In the tilted bifacial configuration, two pyranometers are installed at 15° to monitor the front and rear planes. In the vertical bifacial configuration, two additional pyranometers measure irradiance on the east and west-facing planes. Module temperature is monitored using two Class B PT100 sensors per module. For tilted modules, both sensors are attached to the rear surface, one at the center and the other near the edge. For vertical bifacial modules, temperature is measured at the center of both the front and rear surfaces. Meteorological variables are recorded using a Lufft WS500-

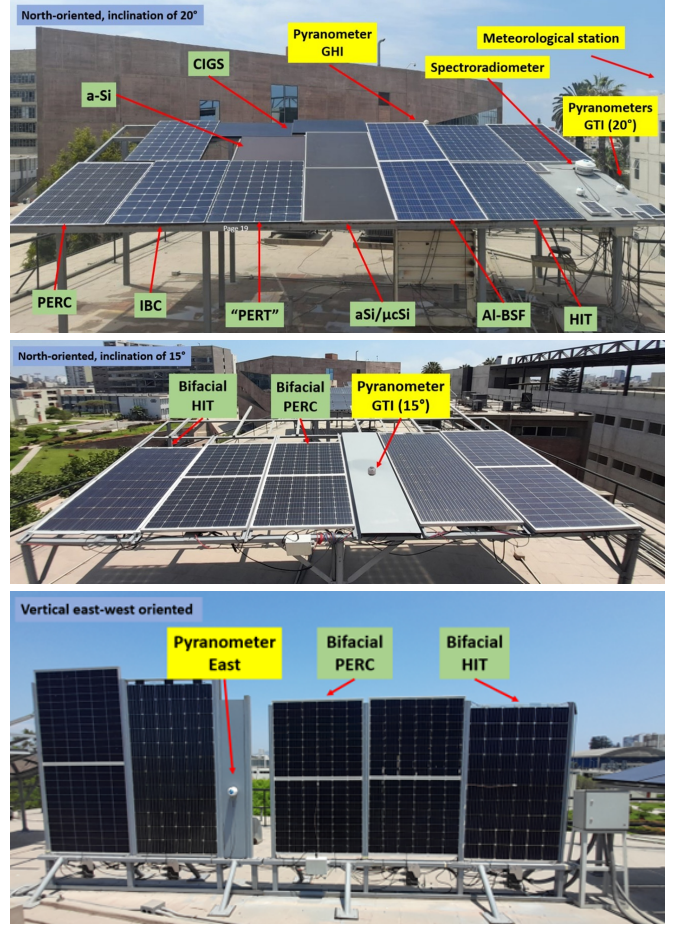


FIGURE 1. (a) Monofacial PV module test station mounted at a 20° north-facing tilt angle. (b) Bifacial PV module test station mounted at a 15° north-facing tilt angle. (c) Vertical bifacial PV module test station with east–west orientation.

UMB weather station, which provides measurements of ambient temperature, relative humidity, atmospheric pressure, air density, wind speed, and wind direction. All PV modules and sensors are cleaned weekly to minimize the effects of dust accumulation.

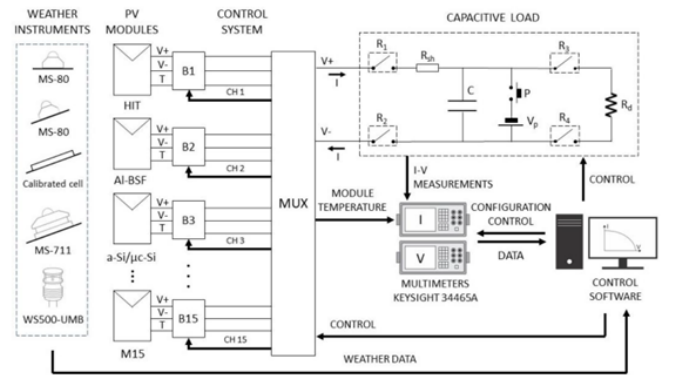


FIGURE 2. Schematic diagram of the PV module monitoring system (reproduced from [12])

The complete measurement system is illustrated in Fig. 2. The PV control unit comprises a 16-channel multiplexer (MUX) and 15 control boxes (B1–B15). Each control box is connected to the positive and negative terminals of a PV module as well as to its corresponding PT100 temperature sensor. Each MUX channel is assigned to a specific control box. When a given channel is activated, the corresponding module is electrically connected, allowing current flow and temperature signal acquisition. Once a PV module is selected, it is connected to the capacitive load circuit. Relays R1 and R2 are closed to initiate capacitor charging, while current and voltage data points are synchronously recorded using two Keysight 34465A digital multimeters. At the same time, the solar spectrum and meteorological variables are measured. After the capacitor is fully charged and the I–V data have been stored, relays R1 and R2 are opened. Subsequently, relays R3 and R4 are closed to discharge the capacitor through resistor Rd, thereby preparing the system for the next measurement cycle.

Electrical parameters were extracted from the measured current–voltage (I–V) curves using the following procedures: The maximum power point (MPP) was identified by computing the instantaneous electrical power as the product of current and voltage and locating its maximum value along the curve. The corresponding voltage and current values were then assigned as the MPP operating point.

The short-circuit current (I_{SC}) was computed using linear regression applied to the initial segment of the I–V curve ($V < \frac{V_{OC}}{6}$). The extrapolated current at zero voltage was taken as the short-circuit current:

$$I = m_1 V + b_1 \Rightarrow I_{SC} = I|_{V=0} = b_1 \quad (1)$$

Similarly, the open-circuit voltage (V_{OC}) was computed by applying linear regression to the final segment ($V > 0.9 \times V_{OC}$) of the I–V curve. The voltage corresponding to zero current was obtained as:

$$I = m_2 V + b_2 \Rightarrow V_{OC} = V|_{I=0} = -\frac{b_2}{m_2} \quad (2)$$

The shunt (R_{sh}) and series (R_s) resistance were estimated as the inverse of the slope of the linear fits used for I_{SC} and V_{OC} extrapolation, respectively.

Figure 3 illustrates a representative I–V curve, highlighting the linear regions used for parameter extrapolation.

Environmental parameters were extracted directly from the corresponding measurement instrument. For irradiance derived from spectroradiometer measurements, the spectral irradiance ($E(\lambda)$) was integrated using the trapezoidal method over the measured range (300–1100 nm).

$$G_{int} = \int_{\lambda_a}^{\lambda_b} E(\lambda) d\lambda \approx \sum_{i=1}^{n-1} \left(\frac{E(\lambda_i) + E(\lambda_{i+1})}{2} \right) \Delta\lambda_i \quad (3)$$

The resulting integrated irradiance was multiplied by a factor of 1.234 to account for the spectral range up to 4000 nm [10]. Average Photon Energy (APE) is computed

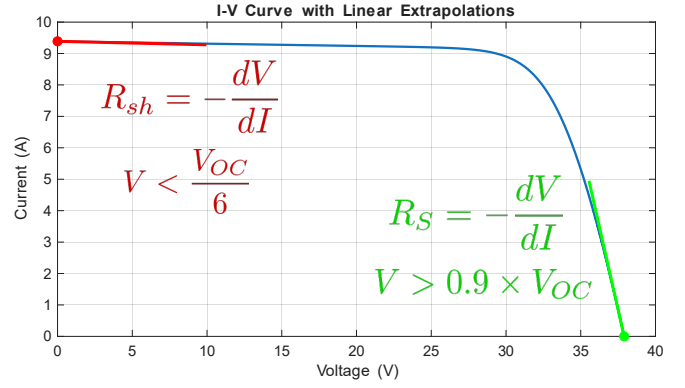


FIGURE 3. I–V Curve showing the linear segments used to extrapolate I_{SC} (red) and V_{OC} (green) and to estimate series and shunt resistances.

using Eq. (4), where h is Planck’s constant, c is the speed of light, and q is the elementary electric charge.

$$APE = \frac{\int_{\lambda_a}^{\lambda_b} E(\lambda) d\lambda}{q \int_{\lambda_a}^{\lambda_b} \phi(\lambda) d\lambda} = \frac{hc}{q} \frac{G_{int}}{\int_{\lambda_a}^{\lambda_b} E(\lambda) \cdot \lambda d\lambda} \quad (4)$$

VALIDATION AND QUALITY

Table 1 shows the specifications for the instruments deployed in the outdoor solar laboratory.

Fig. 4 displays the quantity of missing values for each column in the dataset. The high nullity rate for diffuse irradiance and tilted irradiance at 15° is due to the sensors’ late installation on September 4, 2022. Additionally, the diffuse irradiance sensors had a period of inactivity from May 24, 2023, to December 31, 2025. The rest of the null values correspond to corrupted files or sensor errors. We filtered the data based on the instruments’ measurement limits and physical constraints. After the initial filter stage, Interquartile Range (IQR) filtering was applied to account for extreme outliers. The filtered values were replaced with null values to preserve the rest of the row. Variables with less than 1% of missing values are grouped in the Other bar.

RECORDS AND STORAGE

The presented dataset consists of two files. *PV_dataset.csv* is the time-series log of electrical and environmental variables, while *module_metadata.csv* provides electrical and installation information about the modules presented in the first file. Both files can be related using the *module_name* column.

Table 2 provides a summary of the dataset files.

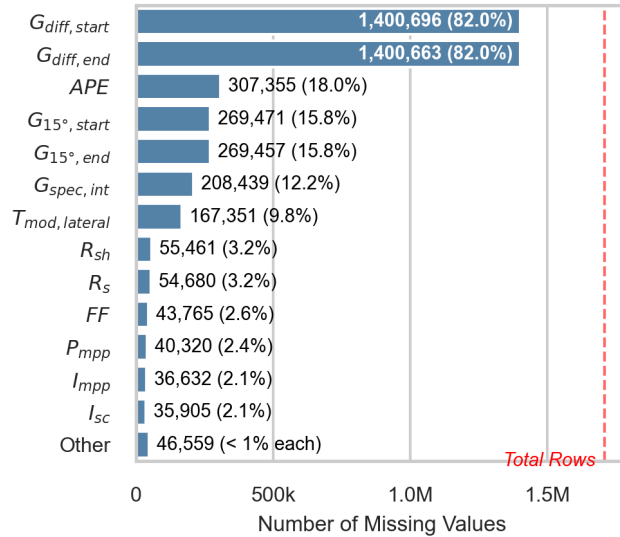
Tables 3 and 4 provide descriptions of the columns present in the corresponding files.

INSIGHTS AND NOTES

Outdoor environmental data is inherently variable. Consequently, the presented dataset captures a wide range of operational conditions, including partial shading, low irradiance, and rapid irradiance fluctuations. Researchers utilizing this dataset must apply task-specific preprocessing or filtering to

TABLE 1. Specifications of the deployed sensors at the Outdoor Photovoltaic Research Laboratory.

Instrument	Measured Variable	Standard / Class	Accuracy & Stability
EKO MS-80S (Pyranometer)	Irradiance (W/m^2)	ISO 9060:2018 Class A	- Non-linearity: $\pm 0.2\%$ - Response time (95%): $< 0.5s$ - Complete zero offset: $\pm 2W/m^2$
EKO MS-711 + RSB-01S (Spectroradiometer)	Spectral Irradiance (W/m^2nm)	NIST Traceable Calibration	- Wavelength Accuracy: $\pm 0.2 nm$ - Optical resolution: $< 7 nm$ - Temperature Response: $< \pm 2\%$ - Wavelength range: $300 - 1100 nm$
WS500-UMB (Smart Weather Sensor)	Environmental Variables	IEC 61724-1:2021	- Temperature: $\pm 0.2 ^\circ C$ - Wind Speed: $\pm 0.3 m/s$ or 3% - Wind Direction: $< 3^\circ$ - Air Pressure: $\pm 0.5 hPa$ - Humidity: $\pm 2\% RH$
RS PRO PT100 (RTD Sensor)	Module Temperature ($^\circ C$)	IEC 60751 Class B	- Stability: $\pm 0.05\%$ - Self-heating: $< 0.5^\circ C/mW$
Keysight 34465A (Digital Multimeter)	Voltage (V) and Current (A)	ISO/IEC 17025 for $k = 2$	- DC Voltage (100V range): $\pm (0.0040\% \text{ reading} + 0.0006\% \text{ range})$ - DC Current: $\pm 0.5\% \text{ reading}$ (75mV/10A shunt)

**FIGURE 4.** Summary of missing data across the dataset. The 25 variables with less than 1% missing data are grouped together.**TABLE 2.** Summary of Data Files

File Name	Entries	Description
PV_dataset.csv	$> 1.7M$	Time-series of electrical and environmental variables over 4 years (2022-2025). 20 modules/configurations.
module_metadata.csv	20	Datasheet and installation info for all modules and configurations.

suit their target applications. For example, training models for Maximum Power Point Tracking (MPPT) typically requires filtering the data to obtain samples with low irradiance variation at the start and end of the recording.

Coefficient of determination (R^2) and Normalized Mean Squared Error (NRMSE) are provided to indicate the quality of the linear fit for extrapolating V_{OC} and I_{SC} , respectively (see Fig. 3). A strong linear fit is evidenced by a high R^2 coefficient and a correspondingly low NRMSE.

Values in the *module_name* column have different suffixes. These correspond to the specific installation configuration:

- ‘-T’, ‘-V’: Tilted or vertical mounting configuration.
- ‘-f’: Bifacial module with the rear side covered (front-side operation only).
- ‘-r’: Bifacial module mounted with the rear side facing upward and the front covered.

The practical utility of the presented dataset for developing AI models has already been demonstrated in prior research. Notably, [14] proposed a shallow neural network for MPPT in combination with a Fuzzy Logic Controller presented in [15]. They trained the network on irradiance, temperature, voltage, and power values from a subset of the present dataset, comprising four modules and one year of data. They achieved low mean square error, with high tracking efficiency of over 98%.

The accompanying repository [16] presents two sample use cases benchmarking Linear Regression (LR), Random Forest Regressor (RF) and Gradient Boosting (GB) models. The PV module temperature predictor estimates module temperature with air temperature, relative humidity, wind

TABLE 3. Description of Columns in PV_dataset.csv

Variable	Unit	Description
module_name	–	Unique identifier for the PV module.
timestamp	–	Local time (UTC-05:00) in YYYY-MM-DD HH:MM:SS.
Vmpp, Impp, Pmpp	V, A, W	Voltage, current, and power at the maximum power point.
Isc, Voc	A, V	Short-circuit current and open-circuit voltage (extrapolated).
FF	%	Fill Factor of the PV module.
Rsh, Rs	Ω	Shunt and series resistance computed from I–V analysis.
R2_Voc	–	R^2 coefficient for linear fit of Voc.
NRMSE_Isc	–	Root Mean Square Error of the linear fit used for Isc extrapolation, normalized by the extrapolated Isc.
V_ini, I_ini	V, A	Initial voltage and current of the traced curve.
G_spec_int	W/m^2	Irradiance from spectral integration.
APE	eV	Average Photon Energy.
Irradiance (G_{**})	W/m^2	Measurements at start/end for: tilted (15° , 20°), horizontal, diffuse, vertical (East, West), and reflected.
module_temp_*	$^\circ C$	Module temperature at center and lateral positions.
air_temp	$^\circ C$	Ambient air temperature.
rel_humidity	%	Measured relative humidity.
air_density	kg/m^3	Computed air density.
abs_pressure	hPa	Absolute air pressure.
wind_*	m/s, $^\circ$	Wind speed and direction.

**Includes: tilt20, tilt15, horiz, diffuse, east, west, refl.

TABLE 4. Description of Columns in modules_metadata.csv

Variable	Unit	Description
module_name	–	Unique identifier for the PV module.
commercial_name	–	Commercial model name of the module.
technology	–	PV technology type (e.g., PERC, HIT, HJT, TOPCon).
bifacial	–	Indicator (yes/no) for bifacial modules.
bifaciality_factor	–	Ratio (0–1) of rear to front performance.
area_m2	m^2	Total physical area of the module.
installation_tilt	$^\circ$	Tilt angle (15° , 20° , or 90° vertical).
*_stc	V, A, W	Datasheet values (V_{oc} , I_{sc} , V_{mpp} , I_{mpp} , P_{mpp}) at Standard Test Conditions.
NOCT_c	$^\circ C$	Nominal Operating Cell Temperature.
temp_coeff_*	$\%/^\circ C$	Temperature coefficients for I_{sc} , V_{oc} , and P_{mpp} .
datasheet_url	–	URL link to the official module datasheet.

speed, and irradiance as features. The GB model achieved a Root Mean Square Error (RMSE) of $4.61^\circ C$ and a R^2 score of 0.835. The PV Power Output model uses module temperature, relative humidity, wind speed, and irradiance to estimate the output power. The best model was GB, achieving an RMSE of $6.81 W$ and a R^2 score of 0.994. This represents an improvement of 67% compared to the formula baseline.

Beyond the demonstrated use cases, the scope of this dataset extends its utility to several broader domains. In photovoltaic engineering, the data can drive the optimization of system installations, such as determining optimal tilt, orientation, and module technology, by analyzing performance across diverse irradiance profiles. Furthermore, the longitudinal nature of the data supports long-term meteorological forecasting and the study of climate change phenomena, including shifts in seasonal intensity. For operational systems, the dataset provides a valuable baseline for predictive maintenance and fault detection, enabling algorithms to identify when measured performance deviates significantly from nominal values. Finally, it serves as a robust benchmark for advancing MPPT and energy management strategies using soft computing techniques, such as artificial neural networks, fuzzy logic, and machine learning-enhanced PID controllers.

SOURCE CODE AND SCRIPTS

The processing pipeline is hosted on GitHub and archived with a permanent DOI to ensure reproducibility:

- Repository URL: <https://github.com/slava-yr/PUCP-Inti-Data>
- DOI: <https://doi.org/10.5281/zenodo.18665161>

This repository includes Jupyter notebooks to extract the raw data and curate it into the presented datasets. These serve as references for the user to understand the decisions taken in the processing of the dataset. The repository includes usage examples as described in the previous section.

The software and libraries used in the development of the dataset are listed in Table 5.

TABLE 5. Software and Library Versions

Software / Library	Version
Python	3.11.14
Pandas	2.3.3
NumPy	2.4.1
Matplotlib	3.10.8
Seaborn	0.13.2
Scikit-learn	1.8.0

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S.Y. curated and analyzed the data, developed the software scripts, and wrote the original draft of the manuscript. M.G. and C.P. conceptualized the methodology and wrote specific sections of the text. M.G. also contributed to the software development. All authors reviewed the manuscript.

The article authors have declared no conflicts of interest.

REFERENCES

- [1] Guillermo Escrivá-Escrivá, “High-Resolution 1-Minute Photovoltaic Power, Temperature and Irradiance Time Series from GEDERlab (UPV),” Jun. 2025. [Online]. Available: <https://doi.org/10.21227/2RYW-2V78>
- [2] A. N. A. Akpolat, “A Photovoltaic System (1 kW) Data for Three Days,” Jan. 2023. [Online]. Available: <https://doi.org/10.21227/EXJT-6M52>
- [3] S. S. Sarang, “Solar Irradiance Data Under Dynamic Weather Conditions (sunny and cloudy days),” 2022. [Online]. Available: <https://doi.org/10.21227/H757-1T06>
- [4] Y. Qiu and Z. Zhou, “2018-2019 PV Generation of a Rooftop Plant in the University of Macau and Weather Report,” Nov. 2020. [Online]. Available: <https://doi.org/10.21227/1KHG-8T55>
- [5] S. Leva, A. Nespoli, S. Pretto, M. Mussetta, and E. Ogliari, “Photovoltaic power and weather parameters,” Sep. 2020. [Online]. Available: <https://doi.org/10.21227/42V0-JZ14>
- [6] M. Sevillano-Bendezú, V. Pleshcheva, B. Calsi, L. Conde, J. Montes-Romero, J. Aguilera, J. De La Casa, and J. Töfflinger, “Assessing the accuracy of analytical methods for extracting parameters of different PV module technologies under clear and cloudy sky conditions,” *Energy Reports*, vol. 12, pp. 4279–4293, Dec. 2024. [Online]. Available: <https://doi.org/10.1016/j.egyr.2024.10.002>
- [7] M. A. Zamalloa-Jara, M. Á. Sevillano-Bendezú, C. Ulbrich, G. Nofuentes, R. Grieseler, and J. A. Töfflinger, “Overirradiance conditions and their impact on the spectral distribution at low- and mid-latitude sites,” *Solar Energy*, vol. 259, pp. 99–106, Jul. 2023. [Online]. Available: <https://doi.org/10.1016/j.solener.2023.05.010>
- [8] M. A. García, G. R. Quispe, M. A. Zamudio, L. A. Conde, J. R. Angulo, A. E. Berastain, and J. A. Töfflinger, “Outdoor I-V characterization of tilted and vertical bifacial PV modules,” *J. Phys.: Conf. Ser.*, vol. 2538, no. 1, p. 012002, Jun. 2023. [Online]. Available: <https://doi.org/10.1088/1742-6596/2538/1/012002>
- [9] M. A. Sevillano-Bendezú, L. A. Conde, J. De La Casa, and J. A. Töfflinger, “Average photon energy assessment based on modelled spectra from the National Solar Radiation Database for Lima, Peru,” *J. Phys.: Conf. Ser.*, vol. 2180, no. 1, p. 012018, Jan. 2022. [Online]. Available: <https://doi.org/10.1088/1742-6596/2180/1/012018>
- [10] L. A. Conde, J. R. Angulo, M. Á. Sevillano-Bendezú, G. Nofuentes, J. A. Töfflinger, and J. De La Casa, “Spectral effects on the energy yield of various photovoltaic technologies in Lima (Peru),” *Energy*, vol. 223, p. 120034, May 2021. [Online]. Available: <https://doi.org/10.1016/j.energy.2021.120034>
- [11] M. Á. Sevillano-Bendezú, L. Á. Conde, M. A. García, J. R. Angulo, J. D. L. Casa, J. M. Ripalda, and J. A. Töfflinger, “Performance loss rate and benchmarking of c-Si and thin-film PV modules considering thermal and spectral effects at a low-latitude site,” *Renewable Energy*, vol. 257, p. 124732, Feb. 2026. [Online]. Available: <https://doi.org/10.1016/j.renene.2025.124732>
- [12] L. A. Conde Mendoza, A. Carhuavilca, R. Perich, J. Montes-Romero, J. Angulo, A. Guerra, E. Muñoz, and J. De La Casa, “Performance Evaluation and Characterization of Different Photovoltaic Technologies Under the Coastal, Desertic Climate Conditions of Lima, Peru,” in *Proceedings of the ISES Solar World Congress 2019*. Santiago, Chile: International Solar Energy Society, 2019, pp. 1–11. [Online]. Available: <https://doi.org/10.18086/swc.2019.16.01>
- [13] J. A. P. López, C. A. Paragua-Macuri, D. A. M. Aucaruri, J. R. A. Abanto, and J. A. Töfflinger, “Classification of solar panel technology and photovoltaic cell status applying machine learning to electroluminescence images,” in *2024 IEEE 22nd Mediterranean Electrotechnical Conference (MELECON)*. Porto, Portugal: IEEE, Jun. 2024, pp. 121–126. [Online]. Available: <https://doi.org/10.1109/MELECON56669.2024.10608551>
- [14] W. R. Ccoñas, D. F. Baca, and C. A. Paragua-Macuri, “Design of an MPPT Algorithm Based on Neural Networks and Fuzzy Logic for Different Photovoltaic Panel Technologies,” in *2025 IEEE XXXII International Conference on Electronics, Electrical Engineering and Computing (INTERCON)*. Arequipa, Peru: IEEE, Aug. 2025, pp. 1–7. [Online]. Available: <https://doi.org/10.1109/INTERCON67304.2025.11244660>
- [15] D. Fernandez and C. Paragua-Macuri, “A Control Algorithm Based on Fuzzy Logic Control for PV Systems in a Non Stable Environment,” in *SoutheastCon 2025*. Concord, NC, USA: IEEE, Mar. 2025, pp. 13–18. [Online]. Available: <https://doi.org/10.1109/SoutheastCon56624.2025.10971588>
- [16] S. Yabar, “Slava-yr/PUCP-Inti-Data: Dataset Processing for IEEE Competition,” Zenodo, Feb. 2026. [Online]. Available: <https://doi.org/10.5281/ZENODO.18665161>